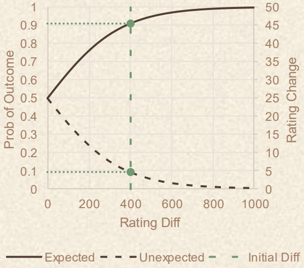
PREDICTING LEVEL OF PLAY IN CHESS

OBJECTIVE

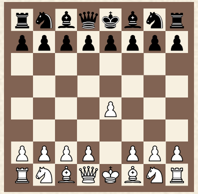
Can a multi-feature approach accurately predict the Elo Rating from a single game?



- Outcome-based
- Limitations in evaluating solo game.
- Inaccurate initial rating.
- Doesn't measure playing strength
- Not suitable for cross-era player comparisons

DATA SOURCE

- Monthly games from Lichess open Database.
- Compressed file of **PGN** data.
- Metadata + Move order seq (SAN).



1. e4 c5 2. Nf3 e6 3. d4 cxd4 4. Nxd4 Nf6 5. Bb5 a6 6. Ba4 Nxe4 7. Qe2 Nf6 8. 0-0 Be7 9. Bg5 0-0 10. Nc3 h6 11. Bh4 b5 12. Bb3 d5 13. Rfe1 Bb7 14. h3 Nc6 15. Nxc6 Bxc6 16. Rad1 Nd7 17. Bxe7 Qxe7 18. a4 Nc5 19. axb5 axb5

DATA & PIPELINE

PGN 250 GB
Unstructured data of games.

GAMES 90M
Played on Lichess.com in February.

PLAYERS 2M
Distinct players who played the games.

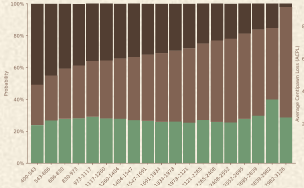
- Parallel processing of batches of games.
- Engine Evaluation to generate features: Eval, W/D/L, Blunders, Mistakes, Inaccuracies, etc.
- Export **tensorized** features to TFRecords for direct ML training.

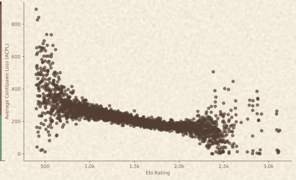
SOURCE
Compressed **.zst** file from Lichess Open Database

BRONZE
Raw unstructured Data **.pgn** file

SILVER
Parsed game data in relational format (SQLite)

GOLD
Aggregated metrics & ML features.

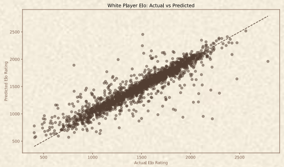


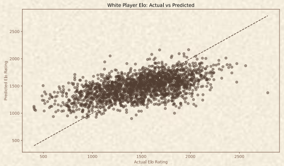


MODELLING

	MODEL:	MLP Regressor			
	HYPER PARAMETERS:	$\alpha = 0.0001$		$L = 100$	
		$g(z) = ReLU$		$Iterr = 5000$	
SIZE	METRICS:	MSE	MAE	R^2	RMSE
40072	TRAIN:	18935.84	83.64	0.86	137.60
10019	TEST:	22324.12	87.84	0.83	149.41

RESULTS





- Predicts Elo with an error of ±87.
- Relied heavily on opponent's Elo.
- Explore sequential models (RNN, CNN, Transformers)